IN CHARACTERISTIC ZERO SMOOTHNESS AT A PRIME, FREENESS OF THE DIFFERENTIALS AND REGULARITY AGREE

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Allowed standard prerequisites (editorial).

- Separability of field extensions in characteristic zero. Used at the start of the counted proof (lines 39052-39053). Gives the equivalence between smoothness and regularity through the separable case.
- Regularity inherited after dividing by a nonzerodivisor. Invoked at the end of the counted proof (lines 39081-39083). Lifts regularity from a quotient along a nonzerodivisor.

Source reference. Original lemma 00DV: sources/stacks-algebra/pinned-original-source.tex, source lines 3833–3833. Navigation pointer for \ref{lemma-NAK} used inside the counted span (line 39072); the argument that consumes it is retained in the span.

Source reference. Original lemma 00TW: sources/stacks-algebra/pinned-original-source.tex, source lines 39001–39001. Navigation pointer for \ref{lemma-characteristic-zero} used inside the counted span (line 39074); the argument that consumes it is retained in the span.

Date: stacks-project:algebra.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

Source reference. Original lemma 00TU: sources/stacks-algebra/pinned-original-source.tex, source lines 38904–38904. Navigation pointer for \ref{lemma-computation-differential} used inside the counted span (line 39068); the argument that consumes it is retained in the span.

Source reference. Original lemma 00RU: sources/stacks-algebra/pinned-original-source.tex, source lines 34320–34320. Navigation pointer for \ref{lemma-differential-seq} used inside the counted span (line 39076); the argument that consumes it is retained in the span.

Source reference. Original lemma 00NU: sources/stacks-algebra/pinned-original-source.tex, source lines 26033–26033. Navigation pointer for \ref{lemma-regular-mod-x} used inside the counted span (line 39083); the argument that consumes it is retained in the span.

Source reference. Original lemma 00TV: sources/stacks-algebra/pinned-original-source.tex, source lines 38957–38957. Navigation pointer for \ref{lemma-separable-smooth} used inside the counted span (line 39053); the argument that consumes it is retained in the span.

Lemma 0.1. *Let k be a field of characteristic 0. Let S be a finite type k-algebra. Let $\mathfrak{q} \subset S$ be a prime. The following are equivalent:*

1. *The algebra S is smooth at $\mathfrak{q}$ over k.*
2. *The $S_\mathfrak{q}$-module $\Omega_{S/k,\mathfrak{q}}$ is (finite) free.*
3. *The ring $S_\mathfrak{q}$ is regular.*

Proof. In characteristic zero any field extension is separable and hence the equivalence of (1) and (3) follows from Lemma 00TV. Also (1) implies (2) by definition of smooth algebras. Assume that $\Omega_{S/k,\mathfrak{q}}$ is free over $S_\mathfrak{q}$. We are going to use the notation and observations made in the proof of Lemma 00TV. So $R = S_\mathfrak{q}$ with maximal ideal $\mathfrak{m}$ and residue field κ. Our goal is to prove R is regular.

If $\mathfrak{m}/\mathfrak{m}^2 = 0$, then $\mathfrak{m} = 0$ and $R \cong \kappa$. Hence R is regular and we win.

If $\mathfrak{m}/\mathfrak{m}^2 \neq 0$, then choose any $f \in \mathfrak{m}$ whose image in $\mathfrak{m}/\mathfrak{m}^2$ is not zero. By Lemma 00TU we see that $\mathrm{d}f$ has nonzero image in $\Omega_{R/k}/\mathfrak{m}\Omega_{R/k}$. By assumption $\Omega_{R/k} = \Omega_{S/k,\mathfrak{q}}$ is finite free and hence by Nakayama's Lemma 00DV we see that $\mathrm{d}f$ generates a direct summand. We apply Lemma 00TW to deduce that f is a nonzerodivisor in R. Furthermore, by Lemma 00RU we get an exact sequence

$$(f)/(f^2) \to \Omega_{R/k} \otimes_R R/fR \to \Omega_{(R/fR)/k} \to 0$$

This implies that $\Omega_{(R/fR)/k}$ is finite free as well. Hence by induction we see that R/fR is a regular local ring. Since $f \in \mathfrak{m}$ was a nonzerodivisor we conclude that R is regular, see Lemma 00NU. □